

2 *Drosophila melanogaster* is a small fruit fly that is often used in research on genetics.

Wild fruit flies normally have dark red eyes due to the presence of a brown pigment, ommochrome, and a bright red pigment, drosopterin.

Fig. 2.1 shows an adult female fruit fly and an adult male fruit fly.

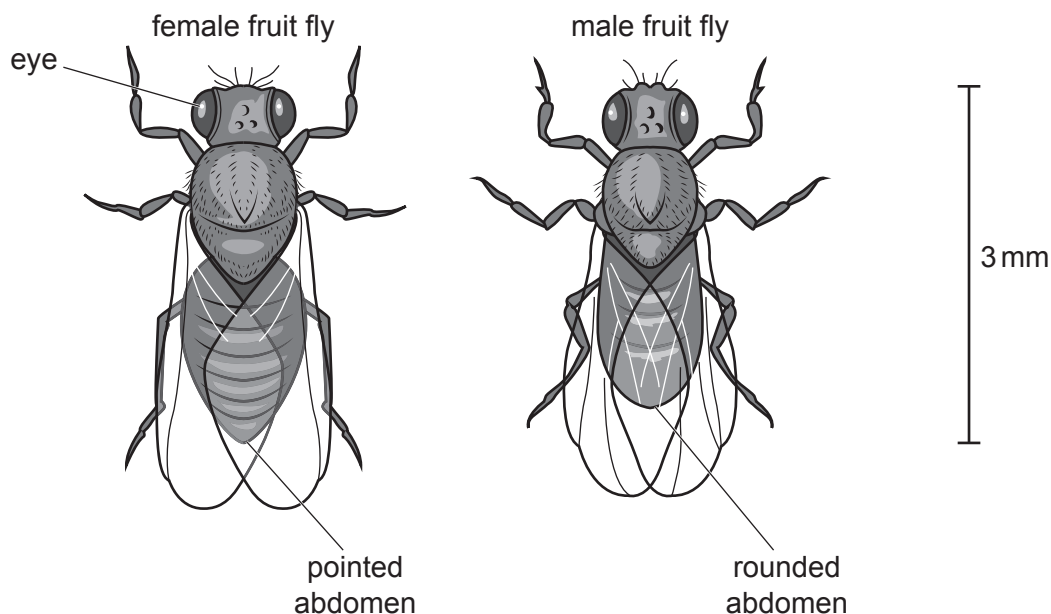


Fig. 2.1

A biologist carried out an investigation to determine the roles of two genes that are involved in the determination of eye colour in adult fruit flies. Both genes have two alleles.

Earlier research by other scientists suggested that:

- the synthesis of the brown pigment ommochrome depends on the gene **B/b**
- the synthesis of the bright red pigment drosopterin depends on the gene **R/r**.

The biologist obtained the parent fruit flies shown in Table 2.1.

Table 2.1

parent fruit flies	genotype	phenotype
female	BBrr	brown eyes
male	bbRR	bright red eyes

The biologist carried out the procedure shown in Fig. 2.2 to cross the parent fruit flies and obtain first generation offspring.

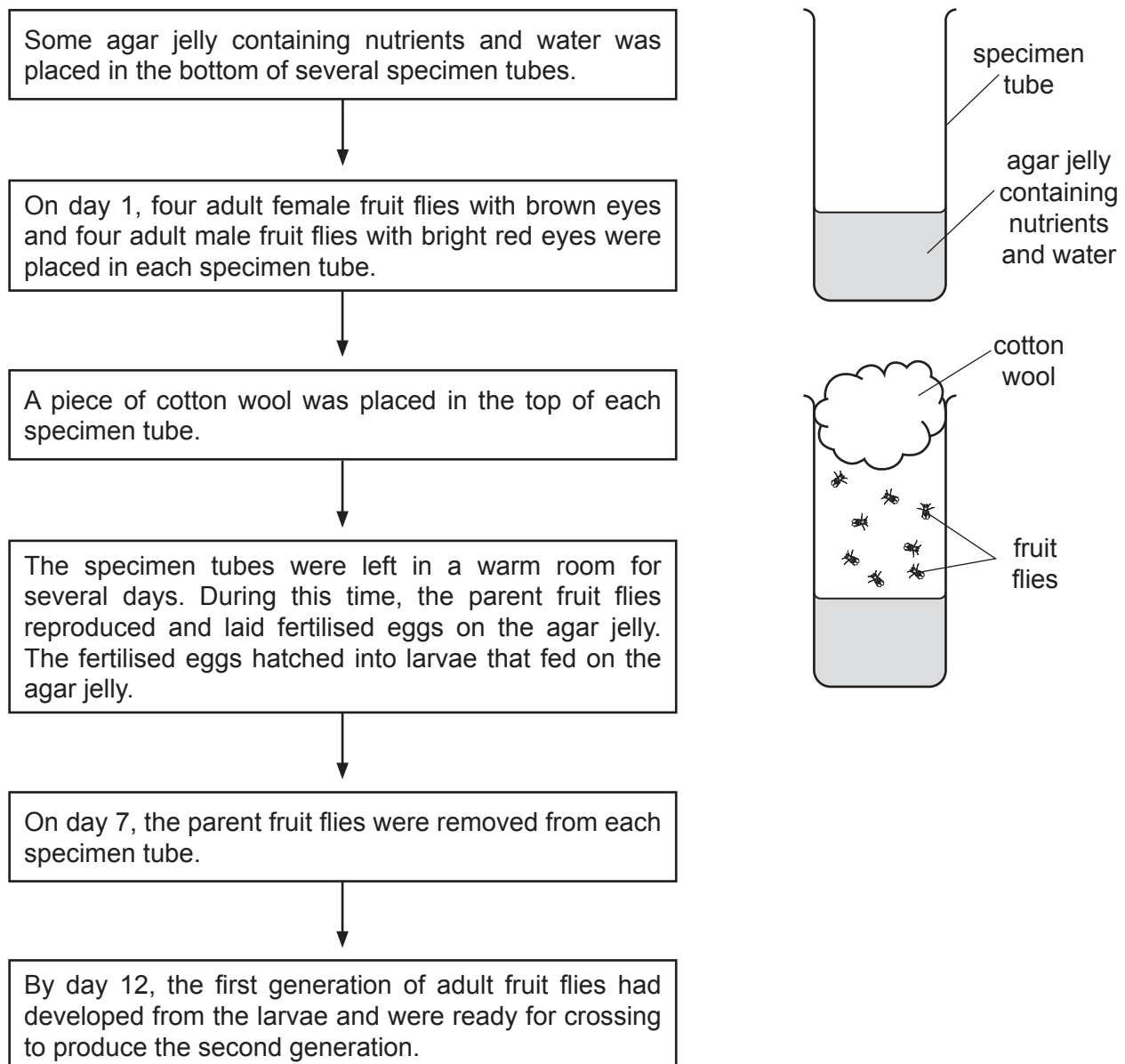


Fig. 2.2

- (a) Suggest a reason why the biologist removed the parent fruit flies from the specimen tubes on day 7.

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The biologist wanted to cross the first generation of fruit flies with one another to produce the second generation of fruit flies.

The first generation of adult fruit flies in the specimen tubes on day 12 were a mixture of females and males and had **not** yet mated.

The biologist crossed the first generation of fruit flies with one another by:

- preparing fresh specimen tubes in which to produce the second generation of fruit flies
- using a chemical to anaesthetise the first generation of adult fruit flies so that they were temporarily unable to move
- separating the adult female fruit flies and adult male fruit flies
- placing four adult female fruit flies **and** four adult male fruit flies into each of the fresh specimen tubes.

- (b) Suggest a method that the biologist could use to separate anaesthetised adult **female** fruit flies and anaesthetised adult **male** fruit flies **and** place four of each into a fresh specimen tube.

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- (c) The biologist expected the cross to result in a phenotypic offspring ratio of 9 : 3 : 3 : 1 in the second generation.

Table 2.2 shows the results of this cross.

Table 2.2

offspring phenotype (second generation)	expected phenotypic ratio	observed number
dark red eyes	9	691
brown eyes	3	260
bright red eyes	3	225
white eyes	1	72
total		1248

The biologist used the chi-squared (χ^2) test to compare the observed and expected results for this cross.

- (i) State the null hypothesis for this chi-squared (χ^2) test.

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 [1]

- (ii) The equation for the calculation of χ^2 is:

$$\chi^2 = \Sigma \frac{(O - E)^2}{E}$$

key to symbols:

O = observed result

E = expected result

Σ = sum of

Complete Table 2.3 to calculate the value of χ^2 for the results of this cross shown in Table 2.2.

Give the value of χ^2 to **four** significant figures.

Table 2.3

offspring phenotype (second generation)	<i>O</i>	<i>E</i>	<i>O</i> – <i>E</i>	$(O - E)^2$	$\frac{(O - E)^2}{E}$
dark red eyes	691				
brown eyes	260				
bright red eyes	225				
white eyes	72				
total	1248				

$\chi^2 = \dots\dots\dots$ [3]

- (iii) The biologist compared the calculated value of χ^2 to the critical values at different probability values shown in Table 2.4.

Table 2.4

degrees of freedom	probability (<i>p</i>)					
	0.95	0.90	0.50	0.10	0.05	0.01
2	0.103	0.211	1.386	4.605	5.991	9.210
3	0.352	0.584	2.366	6.251	7.815	11.345
4	0.711	1.064	3.357	7.779	9.488	13.277

Using Table 2.4 and the calculated value of χ^2 in (c)(ii), explain whether the null hypothesis should be accepted or rejected.

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(d) More recent research has shown that gene **B/b** and gene **R/r** code for polypeptides in carrier proteins. These carrier proteins are found in organelle membranes of the pigment cells of the eyes of adult fruit flies.

- Gene **B/b** codes for a polypeptide in the tryptophan carrier protein. Tryptophan is an amino acid.
- Gene **R/r** codes for a polypeptide in the guanine carrier protein.

Fig. 2.3 shows how the dark red eye colour of wild fruit flies is produced in organelles in pigment cells of the eyes.

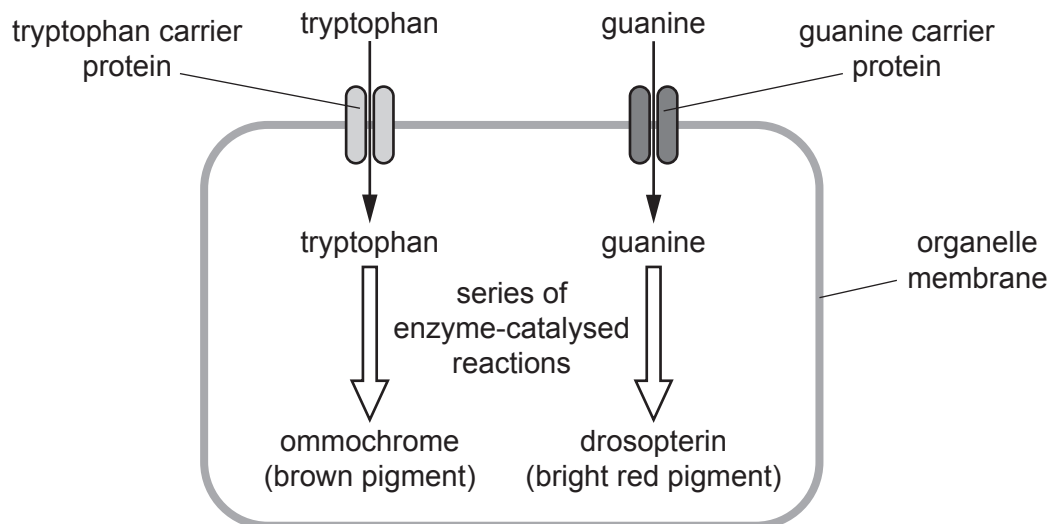


Fig. 2.3

The biologist decided to analyse all the pigments present in the eyes of the second generation of fruit flies.

The biologist started by extracting eye pigments from the adult fruit flies with **dark red eyes** and from the adult fruit flies with **white eyes**. The biologist then added a small volume of each liquid extract to chromatography paper and separated the pigments present by chromatography.

Fig. 2.4 shows the chromatography paper at the end of the procedure when viewed using visible light and ultraviolet light.

Pigment 1 and pigment 3 were visible only when viewed under ultraviolet light. Under visible light, pigment 2 was yellow and pigment 4 was bright red.

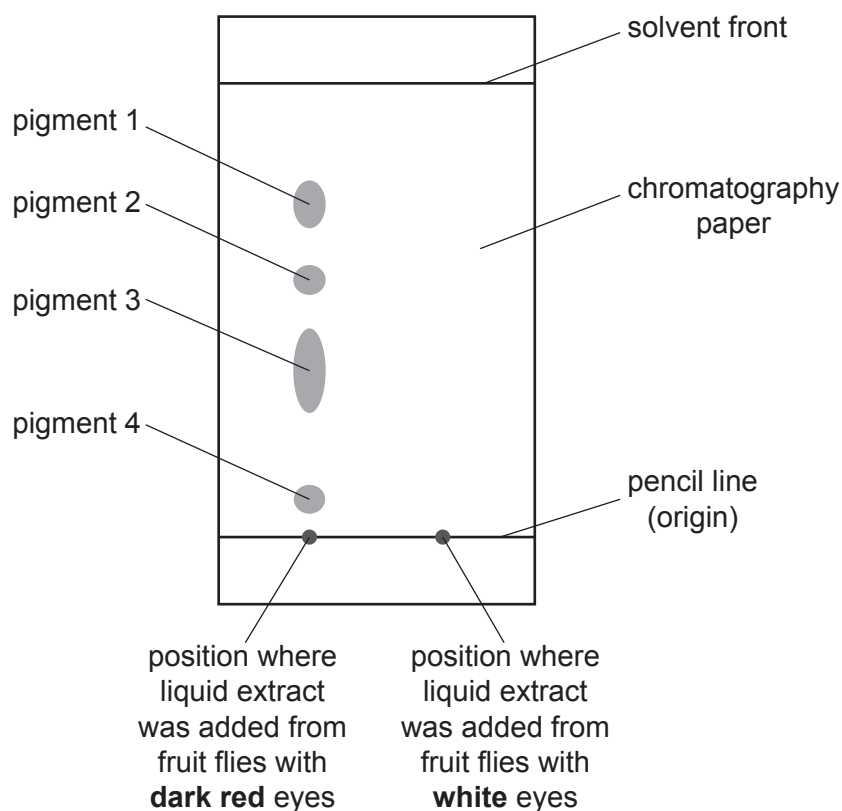


Fig. 2.4

The biologist calculated R_f values for each pigment on the chromatography paper. The biologist used the R_f values to confirm the identity of the pigments using a published source.

(i) The formula for the calculation of R_f is:

$$R_f = \frac{\text{distance moved by pigment}}{\text{distance from origin to solvent front}}$$

When measuring the distance moved by the pigment, the distance to the centre of the pigment should be measured.

Use Fig. 2.4 to calculate the R_f value of **pigment 1**.

$R_f =$ [1]

- (ii) State the conclusions that can be made from the results of the fruit fly breeding experiment in (c) and from the chromatography results.

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[Total: 13]

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